Tameem Zaidat

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Education

Brown University — Providence, RI
Bachelor of Science in Computer Engineering — *Aug. 2023 - May 2027*
Concurrent Master of Science in Computer Science

Work Experience

Digital Exhibit Software Development Intern — May 2025 – Present
Innocent Knowledge — *Providence, RI*

- Developing a full-stack online museum exhibit website using React, TypeScript, Node.js, and Google Firebase, aiming to digitally showcase children's art from conflict zones to a global audience of over **50,000** unique visitors
- Integrating interactive features and multilingual support to foster engagement and accessibility, targeting a 30% increase in user retention through dynamic content delivery

Library System Software Engineering Intern — April 2024 – Aug. 2024
Brown University — *Monroe, MI*

- Led the development of a new library homepage UI using HTML, CSS, and JavaScript, integrating discovery platforms and database access to boost user engagement for **10,000+** Brown University students.
- Optimized multiple full-stack web pages within the library system using LibApps and WordPress, driving content strategy with insights from Google Analytics data.

Fellowships and Awards

Embedded Systems Research Fellow — June 2025 – Present
$1200 *Undergraduate Teaching and Research Award Recipient* — *Providence, RI*

- Individually engineering a real-time embedded system on Raspberry Pi for critical infrastructure monitoring, estimated to reduce equipment downtime by up to 30% through early anomaly detection.
- Implementing hardware-software interfacing via I2C, SPI, and serial protocols for sensor integration.

Deng-Away Web App — 2024
Second Place for Hack for Humanity 2024's ***$3000*** *Seed Grant*

- Developed a full-stack web app using Node.js, HTML, CSS, and JavaScript, to help Indonesian communities identify and report mosquito breeding sites, aiming to reduce dengue prevalence by 15%.
- Designed and implemented a data pipeline to collect, analyze, and visualize crowd-sourced photo submissions on an interactive heat map utilizing some of Google Maps' APIs and locally stored backend data.

Guggenheim Museum Drone Showcase — Sep. 2024 – May 2025
Automatic Coordination of Teams (ACT) Lab Engineering Fellow — *Providence, RI*

- Spearheaded the hardware and software design of lightweight and safe drones to allow for indoor choreographed flights of 60 drones above **1,000,000+** annual visitors of the Guggenheim Museum in NYC.
- Optimized the onboard electronics and creating computer vision planning algorithms using ROS framework

Highlighted Projects

Driving-Day Data App | *Google Firebase, Python, Django* — 2025

- Architected and developed a full-stack web and desktop application to visualize and manage driving-day data for Brown University's Formula SAE team to compete among 120 universities nationwide
- Implemented a robust backend using Google Firebase, Python, and Django to efficiently process, clean, and store live telemetry data, handling approximately 100GB of data weekly.

Annotate | *Swift, SwiftUI* — 2025

- Engineered a macOS application using PDFKit for library management and UserDefaults for data persistence.
- Creating a social annotation system to enable shared reactions and insights on specific text within pages.

Technical Skills

Languages: Java, Python, C#, C, JavaScript, HTML/CSS, SQL, C++, TypeScript, Swift
Frameworks/Tools: Flask, Django, Angular, FastAPI, React, Firebase, AWS, Docker, Node,js, SwiftUI
Relevant Coursework: Data Structures and Algorithms, Discrete Mathematics, Foundations of Artificial Intelligence, Software Engineering, Computer Systems, Cryptographic Systems